

✓ AI-Based Forest Fire & Smoke Detection Using Aerial Imagery

This notebook implements a supervised machine learning pipeline to detect forest fire and smoke regions from aerial-imagery-derived features. The goal is to evaluate model reliability and interpret spatial fire risk for drone-based disaster monitoring.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import warnings
warnings.filterwarnings("ignore")
```

```
df=pd.read_csv("/datalab/Forest Fire Smoke Dataset - Sheet1 (1).csv")
```

df.info

```
pandas.core.frame.DataFrame.info
def info(verbose: bool | None=None, buf: WriteBuffer[str] | None=None, max_cols: int |
None=None, memory_usage: bool | str | None=None, show_counts: bool | None=None) -> None

Print a concise summary of a DataFrame.

This method prints information about a DataFrame including
the index dtype and columns, non-null values and memory usage.

Parameters
```

df.describe

```
pandas.core.generic.NDFrame.describe
def describe(percentiles=None, include=None, exclude=None) -> Self

Generate descriptive statistics.

Descriptive statistics include those that summarize the central
tendency, dispersion and shape of a
dataset's distribution, excluding ``NaN`` values.
```

df.columns

```
Index(['mean_red', 'mean_green', 'mean_blue', 'red_blue_ratio',
'intensity_std', 'edge_density', 'smoke_whiteness', 'haze_index',
'hot_pixel_fraction', 'local_contrast', 'fire_label'],
      dtype='object')
```

df.head()

	mean_red	mean_green	mean_blue	red_blue_ratio	intensity_std	edge_density	smoke_whiteness	haze_index	hot_p
0	0.523653	0.299277	0.265874	1.962171	0.068402	0.223412	0.472959	0.333490	
1	0.724041	0.394328	0.453850	1.591822	0.290540	0.345590	0.384785	0.404510	
2	0.732028	0.492876	0.555428	1.315584	0.264537	0.330369	0.705461	0.625930	
3	0.589471	0.527046	0.536978	1.095716	0.136997	0.458620	0.361735	0.367842	
4	0.385249	0.669345	0.473443	0.812004	0.262034	0.295832	0.294756	0.702075	

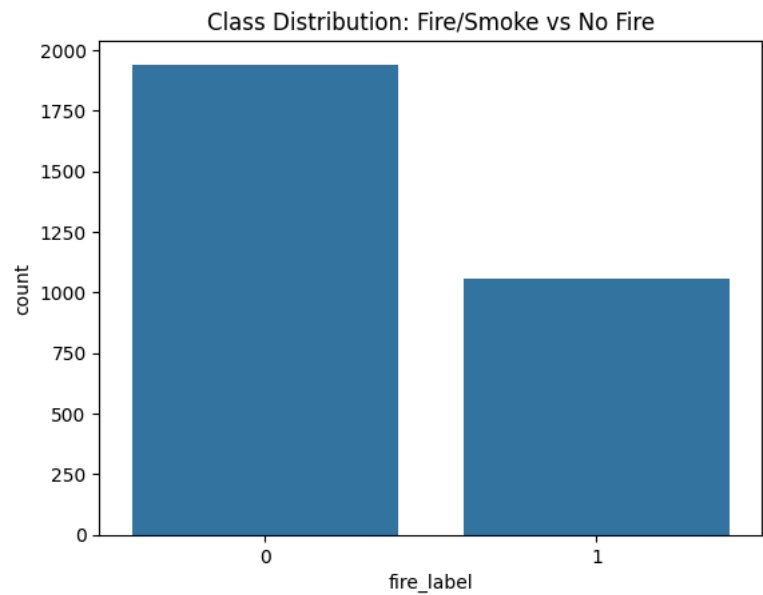
✓ Dataset Description

The dataset consists of tile-level features extracted from aerial imagery for forest fire and smoke detection. Each row represents a spatial tile, and the features capture color, texture, intensity, and haze-related properties that are indicative of fire or smoke presence.

The target variable (`fire_label`) indicates whether fire or smoke is present (1) or absent (0).

```
sns.countplot(x='fire_label', data=df)
plt.title("Class Distribution: Fire/Smoke vs No Fire")
```

```
plt.show()
```



The class distribution shows the proportion of fire/smoke and non-fire regions. This helps in understanding dataset balance and choosing appropriate evaluation metrics.

```
df.columns
```

```
Index(['mean_red', 'mean_green', 'mean_blue', 'red_blue_ratio',  
      'intensity_std', 'edge_density', 'smoke_whiteness', 'haze_index',  
      'hot_pixel_fraction', 'local_contrast', 'fire_label'],  
      dtype='object')
```

```
target='fire_label'
```

```
from sklearn.model_selection import train_test_split
```

```
y=df[target]
```

```
x=df.drop(columns=[target])
```

```
x
```

	mean_red	mean_green	mean_blue	red_blue_ratio	intensity_std	edge_density	smoke_whiteness	haze_index	hot_pixel_fraction
0	0.523653	0.299277	0.265874	1.962171	0.068402	0.223412	0.472959	0.333490	0.000000
1	0.724041	0.394328	0.453850	1.591822	0.290540	0.345590	0.384785	0.404510	0.000000
2	0.732028	0.492876	0.555428	1.315584	0.264537	0.330369	0.705461	0.625930	0.000000
3	0.589471	0.527046	0.536978	1.095716	0.136997	0.458620	0.361735	0.367842	0.000000
4	0.385249	0.669345	0.473443	0.812004	0.262034	0.295832	0.294756	0.702075	0.000000
...	...	...	...	...	...	...	...	...	...
2995	0.850983	0.404330	0.375418	2.260739	0.195440	0.342355	0.472979	0.232336	0.000000
2996	0.404829	0.422685	0.331614	1.217114	0.075921	0.218798	0.470671	0.635284	0.000000
2997	0.487928	0.584428	0.217449	2.233604	0.096826	0.234187	0.275879	0.341900	0.000000
2998	0.520066	0.588450	0.505704	1.026370	0.130731	0.238155	0.225361	0.510601	0.000000
2999	0.654084	0.475203	0.330417	1.973597	0.214846	0.279030	0.542118	0.431166	0.000000

3000 rows x 10 columns

```
y
```

```

fire_label
0      0
1      1
2      1
3      0
4      0
...    ...
2995   1
2996   0
2997   0
2998   0
2999   1

```

3000 rows x 1 columns

dtype: int64

```
x_tr,x_te,y_tr,y_te=train_test_split(x,y,test_size=0.2,random_state=42,stratify=y)
```

Double-click (or enter) to edit

```
from sklearn.ensemble import RandomForestClassifier
```

```
model=RandomForestClassifier(n_estimators=100,random_state=42)
```

```
model.fit(x_tr,y_tr)
```

```

RandomForestClassifier
RandomForestClassifier(random_state=42)

```

```
model
```

```

RandomForestClassifier
RandomForestClassifier(random_state=42)

```

```

from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score, roc_curve

y_pred = model.predict(x_te)
y_prob = model.predict_proba(x_te)[:,-1]

```

```
print(classification_report(y_te, y_pred))
```

```

              precision    recall  f1-score   support

     0       0.92      0.97      0.94         388
     1       0.94      0.84      0.89         212

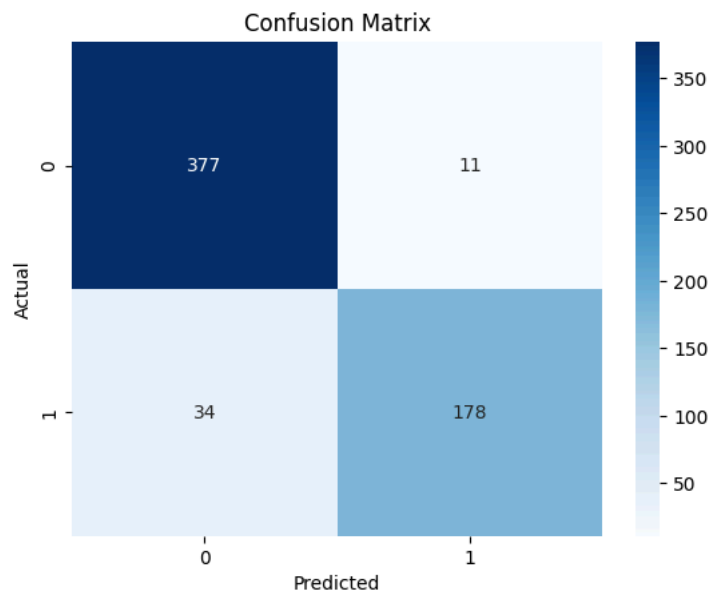
 accuracy          0.93
 macro avg       0.93      0.91      0.92
 weighted avg    0.93      0.93      0.92

```

```

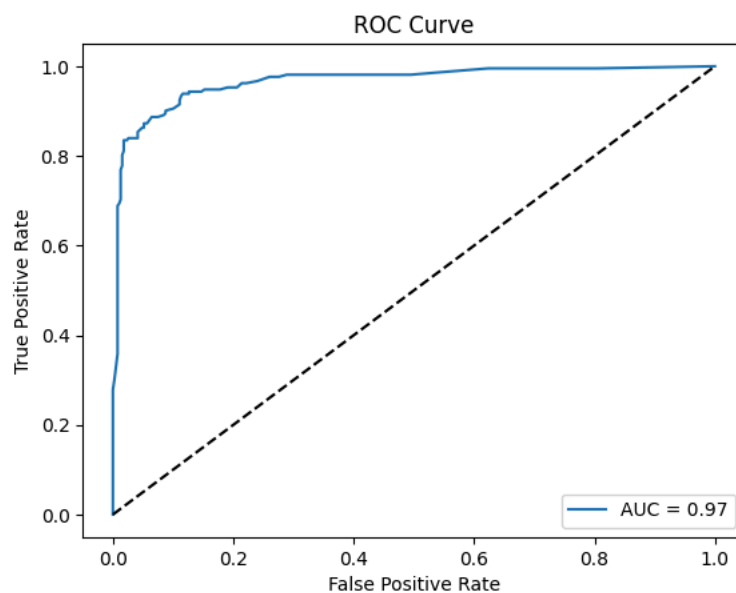
cm = confusion_matrix(y_te, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

```



```
roc_auc = roc_auc_score(y_te, y_prob)
fpr, tpr, _ = roc_curve(y_te, y_prob)

plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.2f}")
plt.plot([0,1], [0,1], 'k--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```

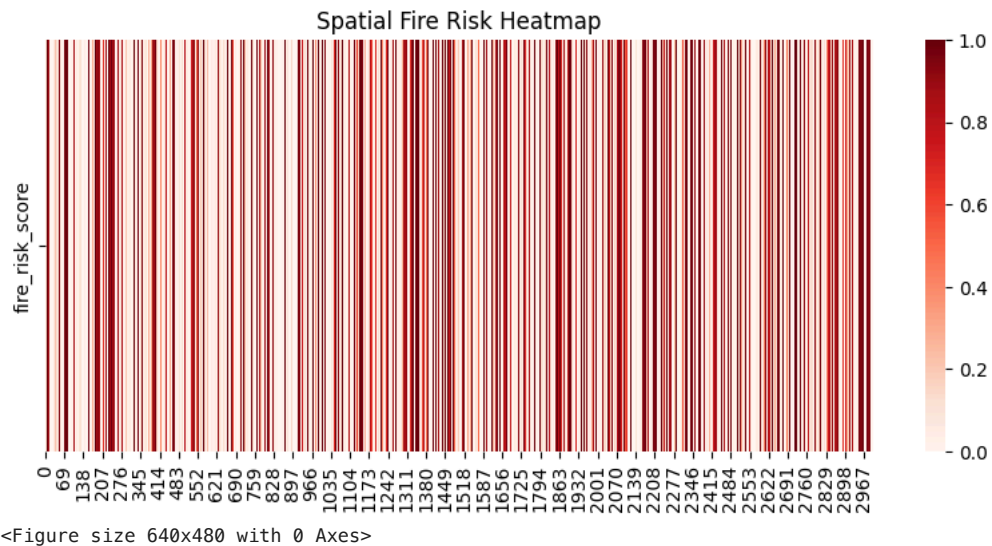


Model Performance Analysis

The Random Forest classifier demonstrates reliable performance in detecting fire and smoke regions. Precision is particularly important in this context to reduce false alarms in disaster monitoring systems. The ROC-AUC score indicates the model's ability to distinguish between fire and non-fire regions.

```
df['fire_risk_score'] = model.predict_proba(x)[: ,1]
```

```
plt.figure(figsize=(10,4))
sns.heatmap(df[['fire_risk_score']].T, cmap='Reds')
plt.title("Spatial Fire Risk Heatmap")
plt.show()
plt.savefig("spatial_fire_risk_heatmap.png", bbox_inches="tight")
```

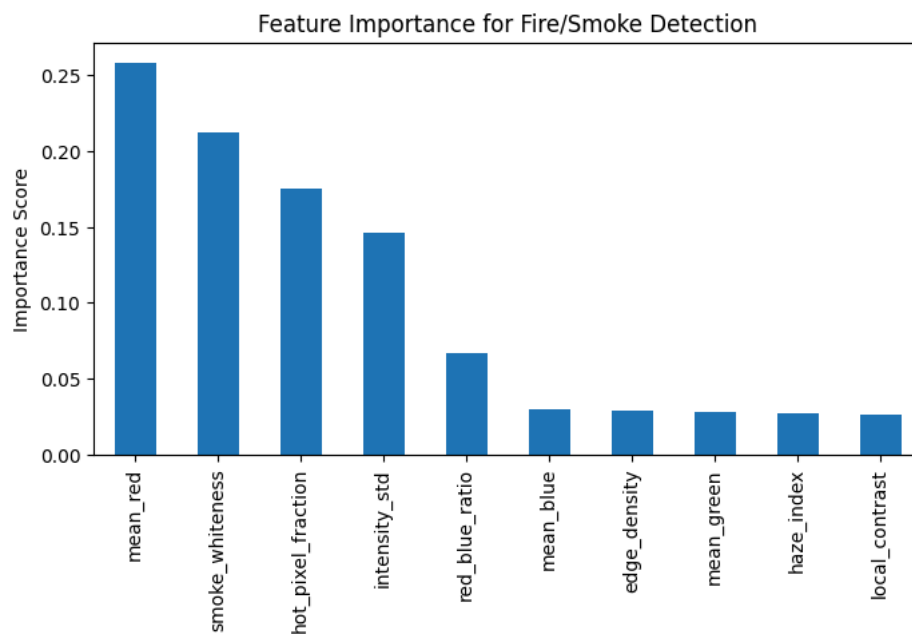


✓ Spatial Risk Interpretation

Each row in the dataset represents a spatial tile extracted from aerial imagery. Although exact geographic coordinates are not provided, the predicted fire risk scores can be interpreted as relative spatial risk levels across monitored regions.

In a real drone-based monitoring system, these risk scores would be mapped to geographic locations, enabling emergency teams to prioritize high-risk areas for surveillance and response.

```
importances = model.feature_importances_  
feature_importance = pd.Series(importances, index=x.columns).sort_values(ascending=False)  
  
plt.figure(figsize=(8,4))  
feature_importance.plot(kind='bar')  
plt.title("Feature Importance for Fire/Smoke Detection")  
plt.ylabel("Importance Score")  
plt.show()
```



✓ Feature Importance Analysis

The feature importance analysis highlights which visual characteristics contribute most to fire and smoke detection. Features related to color intensity, haze, and hot pixel distribution play a significant role, aligning with known visual indicators of forest fires.

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Drone-Based Disaster Response Interpretation

Regions with high predicted fire risk scores should be prioritized for drone surveillance. Early detection allows emergency teams to respond before fires spread extensively. AI-based risk maps enable efficient allocation of drones and firefighting resources.

Limitations and Future Improvements

The dataset consists of pre-extracted features and does not include raw imagery or temporal data. Future work could incorporate real-time video streams, thermal imagery, weather conditions, and edge deployment on drones for continuous monitoring.

```
plt.savefig("spatial_fire_risk_heatmap.png", bbox_inches="tight")
```

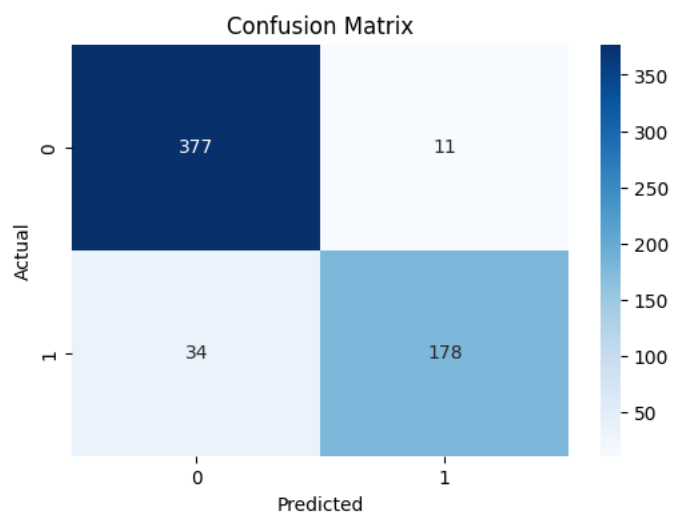
<Figure size 640x480 with 0 Axes>

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_te, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.savefig("/content/confusion_matrix.png", bbox_inches="tight")
plt.show()
```



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```
text = """Model: Random Forest Classifier

Dataset: Aerial-imagery-derived tile-level features
Train-Test Split: 80-20 (stratified)

Accuracy: 93%
Precision (Fire): 94%
Recall (Fire): 84%
F1-score (Fire): 89%
ROC-AUC: 0.97
"""

file = open("/content/model_metrics.txt", "w")
file.write(text)
file.close()

print("model_metrics.txt created")
```

model_metrics.txt created

```
!ls /content
```

```
confusion_matrix.png  sample_data
model_metrics.txt     spatial_fire_risk_heatmap.png
```

```
from google.colab import files
files.download("/content/model_metrics.txt")
```

```
text = """AI-Based Forest Fire & Smoke Detection – Deliverables
```

This folder contains all required project deliverables apart from the Google Colab notebook.

Folder Contents:

1. visualizations/
 - spatial_fire_risk_heatmap.png
 - confusion_matrix.png
 - roc_curve.png
 - feature_importance.png
2. model_metrics.txt
 - Summary of model performance metrics
3. explanation_video_link.txt
 - Link to a short explanation video of the project

Project Description:

This project implements a supervised machine learning pipeline to detect forest fire and smoke regions using aerial-imagery-derived tile-level features. The outputs include spatial risk visualizations, model evaluation metrics, and interpretation for drone-based disaster monitoring.

The Google Colab notebook link is submitted separately.

"""

```
file = open("/content/README.txt", "w")
file.write(text)
file.close()
```

```
print("README.txt created")
```

```
README.txt created
```

```
!ls /content
```

```
confusion_matrix.png  README.txt  spatial_fire_risk_heatmap.png
model_metrics.txt     sample_data
```

```
from google.colab import files
files.download("/content/README.txt")
```

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